3. Sometimes organism may multiply in the skin and produce the toxin which affect the Central Nervous System (CNS) and the effects seen. In the case of *Clostridium tetani* infection, convulsions and paralysis occur due to the production of a powerful toxin.

12.3 Respiratory Tract Infections

With every breath, we inhale several microorganisms and therefore the respiratory system is a major portal of entry for pathogens. In fact, respiratory system infections are the most common type of infections and among the most damaging. Some pathogens that enter via respiratory route can infect other parts of the body, such as skin incase of measles, mumps and rubella.

The upper respiratory system has several anatomical defenses against airborne pathogens. Coarse hairs in the nose, filter large dust particles from the air. The nose is lined with a mucous membrane that contains numerous mucous secreting cells and cilia. The upper portion of the throat also contains a ciliated mucous membrane. The mucous moistens inhaled air and traps dust and microorganisms. The cilia help to remove these particles by moving them towards the mouth for elimination.

12.3.1 Structure of Respiratory Tract

The structure of respiratory tract is divided into two main parts viz: upper respiratory tract (URT) and lower respiratory tract (LRT).

Upper respiratory tract includes mouth, nose, nasal cavity, sinuses, throat or pharynx, epiglottis and larynx.

Lower respiratory tract includes trachea, bronchi, bronchioles, lungs and alveoli (Figure 12.9).

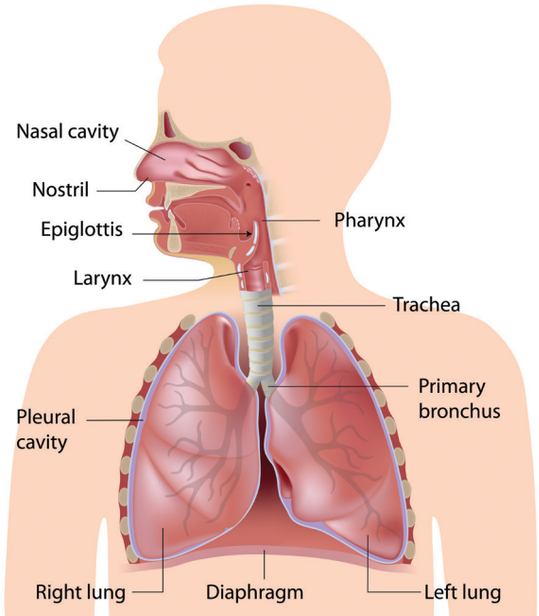


Figure 12.9: Structure of human respiratory tract

12.3.2 Normal Defenses against Infections

Respiratory tract infection are divided into upper respiratory tract (URT) tract infection and lower respiratory tract (LRT) infection. Infection of the respiratory tract are listed in the Table 12.3.

URT: Infections are Sinusitis, Pharyngitis Laryngitis and Epiglotitis

LRT: Infections are Trachiitis, Tracheo bronchitis, Bronchitis, Alveolitis and Pneumonia.

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